

Mobile and Pervasive Computing (CS4123)

F. M. = 30

Time – 2 hrs

Answer all questions

1. a) Write two advantages of PSK over ASK and FSK.
b) Identify two reasons for performing modulation in cellular network.
c) What is block error and how can it be mitigated?
d) How is the capacity of the radio enhanced in cellular network?
e) What is the range of frequency GSM uses in downlink?

[5x2]

2. a) Write steps of operation for establishing a call between two mobile stations.
b) What is the impact of cluster size on capacity and interference in cellular mobile network?
c) Consider a cellular system covers 2310 sq. km and each cell area is 6 sq km. The total allocated channels in the system are 1596. Calculate the system capacity for cluster size 7.

[5+5+5]

3. a) What is channel borrowing ? What are the constraints to implement channel borrowing?
b) What is hand off? Compare between 1st generation and 2nd generation handoff mechanisms.

[2.5x2]

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech(CS) 7th Semester End-Sem Examinations, Nov 2025
Mobile and Pervasive Computing (CS4123)

F. M. = 50

Time – 3 hrs

Answer Question No. 1 and any two from the rest.

1. Define the following terms and state their usage:

- a) Home Agent
- b) Care-of-address
- c) Paging channel
- d) MIMO
- e) Network Slicing

[5x2]

2. a) Define Erlang. How many users can be supported for 0.5% blocking probability for 20 number of trunked channels in a blocked calls cleared system. Assume each user generates 0.1 Erlang of traffic. From Erlang chart it is given that total system load for 0.5% blocking is 11.10.

In the context of CDMA based cellular network, answer the following:

- b) Are the codes (0, 1, 0, 1) and (0, 1, 1, 0) orthogonal? Justify your answer.
- c) Consider two mobile stations A & B want to send data (+1,-1) & (+1,+1) to the base station using the code (-1,+1,-1,+1) & (-1,+1,+1,-1) respectively. Derive the encoded signal received by the base station.
- d) Write the demerits of CDMA.

[4x5]

3. In the context of UMTS, answer the following:

- a) Write algorithmic steps for location tracking and setting up a call originated by land phone.
- b) What are the major functionalities of Node-B?
- c) What is session management? Which protocol is used for packet data communication? Name the relevant parameters to describe such a packet data connection.

[10+5+5]

4. a) What is 5G core network? What are the tasks performed by it?

b) Write the tasks of the following:

- i. NRF
- ii. NSSF
- iii. UDM
- iv. WiFi offloading
- v. Network Aggregation

[10+10]

5. Write short notes on the following:

a) Tunneling in Mobile-IP

b) Pervasive Computing

[10+10]

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
B.TECH CST (7th sem) Mid-Semester EXAMINATION, Sept 2024

Mobile and Pervasive Computing (CS4123)

F. M. = 30

Time – 2 hrs

Answer all questions

1. a) Draw the block diagram of a radio system at both the transmitter and receiver sides. Give a brief description of the tasks of each of the modules in the diagram.
- b) Justify the reason of considering hexagonal cell shape in cellular network design.
- c) Define location tracking. Write about two implementation techniques of location tracking.

[4+3+3]

2. a) What is frequency reuse ratio? Derive the relationship between frequency reuse ratio and signal-to-interference (S/I) ratio.
- b) Consider a GSM TDMA system that accepts $S/I \geq 15$ dB. What should be the compact pattern size N when path loss component $k=3$.

[5+5]

3. Define the following terms and state their usage:

- a) Near-far effect
- b) Umbrella cell approach
- c) GOS
- d) HLR
- e) VLR

[5x2]

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech(CS) 7th Semester End-Sem Examinations, Nov 2024

Mobile and Pervasive Computing (CS4123)

F. M. = 50

Time – 3 hrs

Answer Question No. 1 and any two from the rest.

1. Define the following terms and state their usage:

- a) Channel borrowing
- b) Foreign Agent
- c) MTC
- c) NFV
- e) Mobility-binding table

[5x2]

2. a) Which layer of GSM protocol stack takes care of Radio Resource Management? What are the tasks managed by this module?

b) How do you prove whether a chip code (+1,-1,+1,+1,-1,+1,+1,+1,-1,-1,-1) is orthogonal or not?

c) Consider a network with 20 number of channels per cell and on an average each user makes 3 calls/hr. The average duration of a call is 2 minutes. The offered load from Erlang B Table is 12.03. What is the number of users supported in a cell with 1% blocking?

[10+5+5]

3. Answer the following for the Universal Mobile Telecommunications System.

a) What are the major components of Core network? Briefly state the tasks of each of the components.

b) What roles are played by RNC?

c) Write the names of the traffic types supported in such a system.

[10+5+5]

4. a) Draw the 5G access network architecture showing RAN and Core part.

b) Write advantages of introducing small cells in 5G network.

c) Do a point-wise comparison of Macro, Pico and Femto cell in terms of size, coverage range, cost and deployment.

[5+5+10]

5. Write short notes on the following:

a) Spread Spectrum

b) Context-Aware Computing and Applications

[10+10]

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
B.TECH CST (7th sem) Mid-Semester EXAMINATION, Sept 2023

Mobile and Pervasive Computing (CS4123)

F. M. = 30

Time – 2 hrs

Answer all questions

1. i) Which of the following is the process performed at receiver end in mobile communication?

A) Modulation B) Demodulation C) Decoding D) Both B) & C)

ii) Which of the following stores the data related to the user?

A) SIM B) HLR C) VLR D) AUC

iii) Which type of multiplexing enables to use of the whole bandwidth simultaneously?

A) FDMA B) CDMA C) TDMA D) None of these

iv) Select the incorrect statement about TDMA from the following options:

A) High transmission rate B) Discontinuous data transmission
C) Single carrier frequency for single user D) All of these

v) How is the capacity of the radio enhanced in cellular network?

A) By increasing the total base stations and by channel reuse
B) By increasing the spectrum of the radio C) Both of these D) None of these

vi) Which technique is used to increase cell capacity using directional antenna?

(A) Cell Splitting
(B) Coverage zone approaches
(C) Cell Sectoring
(D) Cell Sectoring and Cell Splitting both

vii) In CDMA, transmission power of mobile devices needs to be adjusted to mitigate the following problem

A) Hidden station problem B) Exposed station problem C) Near-far problem
D) All of these

viii) Primary GSM uses uplink frequency in the range between _____.

A) 890- 915 MHz B) 935-960 MHz C) 880-915 MHz D) 925-960 MHz

ix) Which of the following needs to be estimated for allocating RF number of channels to meet GOS.

A) Cost B) Capacity C) SNR D) All the above

x) Visitor location register is integrated with which of the following?

A) MSC B) HLR C) PSTN D) All these

[10x1=10]

2. a) Write steps of operation for setting a call from mobile to mobile.

b) What roles the channels FVC, RVC, FCC and RCC play in cellular mobile network?

[5+5]

3. a) Describe different channel assignment strategies in 2G cellular network.

b) i) In a cellular system 40 MHz bandwidth is allocated. The system uses two 20 KHz simplex channels to provide full duplex voice and control channels. How many number of channels each network cell may get for a 12-cell reuse system?

ii) Consider the system covers 2310 sq. km and each cell area is 6 sq km. Calculate the system capacity.

[5+5]

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech(CS) 7th Semester Final Examinations, Nov-Dec 2023
Mobile and Pervasive Computing (CS4123)

F. M. = 50

Time – 3 hrs

Answer Question No. 1 and any two from the rest.

1. Define the following terms and state their usage:

- a) Softer-handover
- b) HLR
- c) User plane
- d) Control plane
- e) MIMO

[5x2]

2. a) Define GOS. Write assumptions (if any) in case of blocked calls cleared trunking system.

b) Define Erlang. How many users can be supported for 0.5% blocking probability for 20 number of trunked channels in a blocked calls cleared system. Assume each user generates 0.1 Erlang traffic. From Erlang chart it is given that total system load for 0.5% blocking is 11.10.

c) Write the algorithmic steps of CSMA/CA in WLAN.

d) What is tunneling in Mobile-IP?

[4x5]

3. a) What are the major components of UTRAN? Briefly state the tasks of each of the components.

b) In UMTS signaling protocol stack, what roles are played by 'Call management' and 'Mobility management'?

[15+5]

4. a) What is 5G core network? What are the tasks performed by it?

b) Write the tasks of the following functions:

- i. NRF
- ii. NSSF
- iii. PCF
- iv. UDM

[10+10]

5. Write short notes on the following:

a) Career aggregation

b) Pervasive Computing

[10+10]

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
B.TECH (7th sem) Mid-Semester EXAMINATION, Sept 2022

Mobile and Pervasive Computing (CS4123)

F. M. = 30

Time – 2 hrs

Answer all questions

1. A. Which of the following is a fundamental principle of wireless communication?
(a) Electromagnetic waves (b) Microwaves (c) Both A and B (d) None of the above
- B. What is transmitted with the call initiation request during a mobile call.
(a) MIN (b) ESN (c) Both a) & b) (d) None of the above
- C. Which technique is used to increase cell capacity using directional antenna?
(a) Cell Splitting (b) Coverage zone approaches (c) Cell Sectoring (d) Cell Sectoring and Cell Splitting both
- D. How is the capacity of the radio enhanced in cellular?
(a) By increasing the total base stations and by channel reuse (b) By increasing the spectrum of the radio (c) Both of these (d) None of these
- E. In GSM, handover is taken care of by the following sub-layer of the Network layer
(a) Call Management (b) Radio Resource Management (c) Mobility Management (d) None of these
- F. Which one of the following enables us to use the entire bandwidth simultaneously?
(a) TDMA (b) CDMA (c) FDMA (d) All of the above
- G. Which of the following can be considered as the advantage of using frequency reuse?
(a) The same spectrum can be allocated to the other networks
(b) Only a limited spectrum is required
(c) Increase capacity
(d) All of the above

H. How many sub-systems are there in Global Systems for Mobile (GSM)?

(a) 4 (b) 3 (c) 2 (d) None of the above

I. Name the sub-systems mentioned in 1.H.

J. **Mobile IP** is a communication protocol that allows the users to move from one network to another with

(a) One IP address (b) Two IP address (c) No IP address (d) All the above may be possible

[10x1=10]

2. a) Why cell shapes of the cellular network architecture is generally considered as hexagon?

b) Define location tracking. Write about two implementation techniques of location tracking.

[5+5]

3. a) Define GOS. Mention assumptions (if any) in case of blocked calls cleared trunking system.

b) Define Erlang. How many users can be supported for 0.5% blocking probability for 20 number of trunked channels in a blocked calls cleared system. Assume each user generates 0.1 Erlang of traffic. From Erlang chart it is given that total system load for 0.5% blocking is 11.10.

[5+5]

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech(CS) 7th Semester Final Examinations, Dec 2022
Mobile and Pervasive Computing (CS4123)

F. M. = 50

Time – 3 hrs

Answer Question No. 1 and any two from the rest.

1. Define the following terms and state their usage:

- a) Soft-handover
- b) VLR
- c) NFV
- d) SDN
- e) Network slicing

[5x2]

2. a) Draw a diagram of UMTS architecture showing its major components. Briefly describe the task of each component.

b) What is close loop power control in CDMA based cellular network system? Explain in the context of 3G UMTS.

[10+10]

3. a) Are the CDMA codes (1, 0) and (1, 1) orthogonal? Justify your answer.

b) In a CDMA system, consider two Mobile stations transmit data-a (1,0) and data-b (1,1) respectively. The system assigns two unique and orthogonal codes – code-a = (0, 1, 0, 1), code-b = (0, 1, 1, 0). Both the MSs spread respective signal using their unique key as chip sequence. For convenience binary 0 is assumed as -1, and binary 1 as +1.

What will be the resulting signals (encoded data-a & data-b) received at base station?

c) Write advantages of using small cells in 5G over traditional cell towers.

d) What type of cell is recommended to get 5G coverage in each of the following?

- i. Shopping mall
- ii. Residential house

Justify your answer.

[4x5=20]

4. a) Explain 5G service based system architecture.

b) Write the tasks of the following functions:

i. NSSF

ii. AMF

iii. SMF

iv. AUSF

[10+10]

5. Write short notes on the following:

a) Mobile-IP

b) Context-Aware Computing and Applications

[10+10]